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Developing Linguistic Constructs of Text Readability Using Natural Language Processing

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ABSTRACT

Purpose: Understanding what makes a text difficult to read is an important element of literacy instruction, materials development, and research into text readability. Readability formulas are often used to quantify text readability. However, readability formulas often focus on a relatively limited number of text features that inform the reading process and/or use single features to measure complex language constructs under the presumption that related features do not interact with one another. As a result, the formulas can be difficult to interpret by practitioners.

Method: A principal component analysis was used to combine text features from advanced natural language processing tools into theoretically-inspired language constructs related to text comprehension.

Result: Results indicate that text features aggregated into seven components related to informational text, syntactic constructions, noun phrase complexity, lexical ease, function word ease, narrativity, and lexical variety. These components predicted 44% of the variance in readability scores reported in a large readability corpus.

Conclusion: The results provide evidence that individual linguistic features related to text readability can be combined into reading components that are strong predictors of text comprehension. Practically, these components should be more easily interpretable allowing for more actionable feedback and use by practitioners.

INTRODUCTION

Understanding what makes a text difficult to read is an important element of literacy instruction, materials development, and research into text readability. The Common Core State Standards (Common Core State Standards Initiative, 2010), often used in the United States, detail three factors that influence text readability: reader and task factors, qualitative text complexity factors, and quantitative text complexity factors. Research has indicated that much of text readability can be explained by readers' background knowledge or reading ability (O'Reilly et al., 2019; Wang et al., 2017). Qualitative factors as described in the CCSS dimensions are difficult to measure because they relate to variations in text that an attentive reader can measure that influence readability. However, quantifying text difficulty using language features in text and using these features to predict text readability has a long research history. For instance, research indicates that a text will likely be more difficult to read if it contains more difficult words, more complex sentence structures, and fewer cohesive features that bind sentences and paragraphs together (Just & Carpenter, 1980; Snow, 2002). Studies have shown that lexical features including imageability and concreteness (Richardson, 1975), word familiarity and frequency (Howes & Solomon, 1951), sound and spelling relationships between

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words (Juel & Solso, 1981; Mesmer, 2005), and semantics (Collins-Thompson, 2014; Mesmer et al., 2012) influence word reading. Effective reading also results from less complex syntactic structures (Cunningham et al., 2005; Mesmer et al., 2012) and increased connections across words, structures, and concepts through discourse structures (Givón, 1995; Mesmer et al., 2012). Additionally, pragmatic features in a text like sentiment can help readers maintain engagement and motivation (Collins-Thompson, 2014). Lastly, language features can combine as a function of genre, leading to texts that are easier to read. For instance, research indicates that narrative texts are easier to comprehend compared to informative or expository texts (Best et al., 2008; Sáenz & Fuchs, 2002).

Knowing the effect of language features on text readability, researchers have developed hundreds of readability formulas over the past 80 years (Crossley, Skalicky, et al., 2017). These readability formulas quantify linguistic features in the text that act as proxies for language features known to influence the reading process. These features are then used in statistical models to predict text readability. For instance, Flesch Reading Ease (Flesch, 1948) is an early readability formula that counted the number of syllables in words as a proxy for word complexity (i.e., lexical sophistication) and the number of words per sentence as a proxy for sentence complexity (i.e., syntactic complexity). Readability formulas have played an important role in a variety of learning environments by allowing teachers, administrators, and materials developers to better match readers with texts that complement their reading level, which is a critical component of learning to read (Allington, 2005; Stanovich, 1985; Wolfe et al., 1998).

However, there are many existing problems with available readability formulas. The first and potentially the most important concern is that readability formulas often focus on a relatively limited number of features that inform the reading process. Almost all popular readability formulas developed before the 21st century, including Flesch–Kincaid Grade Level (Kincaid et al., 1975), relied on proxies for lexical sophistication and syntactic complexity and ignored language features such as cohesion, semantics, morphology, and pragmatics. Additionally, most readability formulas use single linguistic indices to measure more complex language constructs and their relationships to text readability. For instance, the New Dale Chall formula (Chall & Dale, 1995) counts the incidence of the 3,000 most frequent words in the English language and uses this incident count as a proxy for lexical sophistication. However, a theoretically inspired construct of lexical sophistication related to reading comprehension would be much more robust and include features related to word processing like grapheme to phoneme mapping, word rhyming, and neighborhood effects to name but a few. Commonly used readability formulas developed in industry, like Lexile (Smith et al., 1989), are also problematic because very little information is available about the linguistic features contained with the formulas or their reliability. What little is available indicates that they also rely on surface-level feature like those measured in the Flesch–Kincaid Grade Level formula (Graesser et al., 2011; Sheehan et al., 2015).

More advanced readability formulas often include multiple theoretically inspired features that relate to single language construct. However, these formulas usually have an implicit assumption that these features do not interact with one another and instead work individually to explain text difficulty. For instance, the Crowdsourced Algorithm of Reading Comprehension (CAREC, Crossley, Kyle, et al., 2019) includes 13 theoretically inspired linguistic features related to lexical sophistication (word age of acquisition, frequency, imageability, character entropy), n-gram features (bi-gram range, tri-gram proportion scores, trigram type–token ratio), and cohesion (lexical overlap at the paragraph and sentence level). The nature of the linear regression used to develop the formula, however, treats each feature individually regardless of potential links among them. Treating related features individually is problematic because the importance of larger language constructs and their overall effect on reading comprehension are overlooked. Additionally, readability formulas that are based on individual features and not larger language constructs are more difficult to interpret by practitioners and users including materials developers, teachers, administrators, and students.

There are a few exceptions to an individual feature approach for understanding text readability. These exceptions focus on developing aggregated factors of readability from text features using principal component analyses (PCA, Graesser et al., 2011; Sheehan et al., 2014). For instance, Graesser et al. (2011) extracted ~50 linguistic features from over 35,000 texts in the Touchstone

Applied Science Associates (TASA) corpus using the Coh-Metrix natural language processing (NLP) tool (Graesser et al., 2004; McNamara et al., 2014).[1] Using principal component analysis, these linguistic features were combined into eight components related to narrativity, syntactic simplicity, word concreteness, referential cohesion, deep cohesion, verb overlap, connectives, and temporal cohesion. Visualizations of the components indicated that they aligned with grade-level expectations and genre differences in the texts. Graesser et al. (2011) suggested that the components could be used to tailor text assignment adaptively to students to better match individual reading profiles and address reading deficits and skills. These components were later incorporated into the Test Ease and Readability Assessor (TERA, Jackson et al., 2016).

While the TERA language components developed by Graesser et al. (2011) visually relate to text readability, their relationship to readability is not clear. Primarily this is because the TASA corpus used to develop the components has no reading criteria assigned to each text. Instead, each text in the TASA corpus is assigned a presumed grade level based on a Degree of Reading Power (DRP, Koslin et al., 1987) score. DRP scores are based on weak proxies of language constructs including the number of letters per word and the number of words per sentence (Dufty et al., 2006), so the grade levels assigned to each text in TASA are weak estimates of actual readability. In a follow-up study, Graesser et al. (2014) did examine correlations between the TERA language components and Flesch–Kincaid Grade Level and Lexile scores reported on the TASA corpus. Graesser et al. found strong correlations between the TERA language components narrativity and syntactic ease with Flesch–Kincaid Grade Level and Lexile scores ($r < .50$), but weak or no correlations with TERA language components related to word concreteness and cohesion.

The current study builds on Graesser et al. (2011) by examining newly developed and theoretically inspired linguistic components related to reading. We develop reading components based on independent reading samples taken from a diverse corpus of texts where each excerpt has a unique readability score. In addition, we use a suite of NLP tools to assess a wider breadth of linguistic features in the text from which to derive reading components. These tools assess many linguistic features unavailable when the Graesser et al. (2011) study was completed including variables related to sentiment, cognition, syntactic constructions, lexical diversity, and morphology. For example, a number of lexical diversity calculations that control for text length are now available. Similarly, newer features that capture the morphological complexity of words including morpheme counts, variety, and complexity are now available as are robust measure of sentiment (e.g., the NRC Emotion Lexicon, Mohammad & Turney, 2013) and cognition (i.e., word vectors for semantic constructs such as *thinking, knowing*, and *perceiving*). We use these features along with classic NLP features to derive reading components and then model readability using the derived component scores. We compare these readability models to models developed using the TERA components. The goal of the study is to examine how well newly developed component scores overlap with linguistic constructs known to influence reading and to assess the performance of these components in predicting text readability. We also examine overlap between the reading components developed in this study and those reported by TERA. The research questions that guide this study are:

(1) What linguistic components are derived from reading excerpts?
(2) Are these components similar to those reported by TERA?
(3) How well do these components predict text readability when compared to TERA components?

METHODS

CLEAR corpus

The reading excerpts used in this study come from the CommonLit Ease of Readability (CLEAR) corpus (Crossley et al., 2023). The reading excerpts comprise 4,724 reading passages related to a variety of domains including literature, journalism, the arts, science, history, and

social studies. The reading excerpts are a mixture of literary/imaginative ($n = 2{,}420$) and informational/factual ($n = 2{,}340$) texts. The length of each reading text was restricted to 140–200 words that represent one coherent idea. Exceptions were limited by author, and no two excerpts were sampled from the same text. While extracting the reading excerpts, preference was given to collecting excerpts from the middle of texts ($n = 3{,}470$) although excerpts were extracted from the beginning of the texts ($n = 1{,}024$), and the end of texts ($n = 108$). A small number of excerpts were entire texts ($n = 122$). Excerpts were sampled from text written from approximately 1850 to 2020 with a noticeable gap from around ~1950, where copyright was enforced, to ~2000 when reading excerpts became freely available on the internet. The reading texts consisted of only prose (i.e., no poetry was included). While collecting the reading excerpts, mature/sensitive topics (e.g. alcohol consumption, violence, implicit or explicit racism) that are not suitable for K-12 students were avoided. For a full description of the corpus, its collection, the scoring mechanisms, and reliability, please see Crossley et al. (2023). The CLEAR corpus is also available to download for free at https://www.commonlit.org/blog/introducing-the-clear-corpus-an-open-dataset-to-advance-research-28ff8cfea84a/.

Text readability in the CLEAR corpus

Text readability scores for each individual excerpt were derived from teachers' judgments of how readable a text was in comparison to other texts. This approach is related to the qualitative factor of the CCSS dimensions (2010) in which attentive readers can measure variations in text that influence readability that may not be captured by reader and task factors or quantitative text complexity factors.

The readability scores extracted from the teachers' judgments were based on Bradley–Terry (BT, R. A. Bradley & Terry, 1952) coefficient scores, an approach used in previous studies (Crossley, Kyle, et al., 2017, 2019; De Clercq et al., 2014). BT coefficient scores were the product of pairwise comparison judgments of the reading passages collected from 1,116 middle and high school teachers. The participating teachers were randomly assigned to read pairs of reading passages. For each pair, teachers were asked to judge which passage was easier to read for their students. The teachers made 111,347 judgments of text readability and each text, on average, was read by ~45 teachers.

Rankings (of pairs of texts) from teachers were collected and Bradley–Terry pairwise comparison models were conducted on the rankings to identify a reading ease score for each excerpt using the Bradeley-Terry2 package available in R (Turner & Firth, 2020). A Bradley–Terry model describes the probabilities of the possible outcomes when items are judged against one another in pairs (see Equation 1). The Bradley–Terry model ranks documents by difficulty based on each excerpt's probability to be easier than other excerpts. The model creates a maximum likelihood estimate which iteratively converges toward a unique maximum that defines the ranking of the excerpts (i.e., the easiest texts have the highest probability and the highest scores). After computation, the Bradley–Terry model provides a coefficient for each text along with a standard error. These coefficients were continuous variables that ranged from ~ −3.5 to ~1.5 and were normally distributed. The coefficients represented the ease of reading each text (i.e., a reading ease score). The coefficients were not scaled to grade level, but they did strongly correlate with Flesch–Kincaid Grade Level scores for the texts ($r = .52$, Crossley et al., 2023).

Crossley et al. (2023) developed a readability model using the reading ease scores reported in the CLEAR corpus using 28 linguistic variables related to lexical sophistication, text cohesion, syntactic complexity, lexical diversity, and sentiment and cognition. Their model explained 51% of the variance in the scores. The strongest predictors were related to word decoding, phrasal complexity, and sentiment and cognition terms. Significant, but less predictive features were related to text cohesion.

TERA components

We used the Test Ease and Readability Assessor (TERA, Jackson et al., 2016) to compute the component scores reported in Graesser et al. (2011). We used a desktop version of TERA available internally to researchers at Arizona State University that provides scores for each of the eight components reported in Graesser et al. (i.e., narrativity, syntactic simplicity, word concreteness, referential cohesion, deep cohesion, verb overlap, connectives, and temporal cohesion). There is a web version for TERA available at https://soletlab.adaptiveliteracy.com:8443/.

NLP tools for new component development

We used a number of newer NLP tools from the Suite of Automatic Linguistic Analysis Tools (SALAT; Crossley & Kyle, 2018 to derive theoretically inspired linguistic features from the CLEAR excerpts. These linguistic features build on those reported in Coh-Metrix (Graesser et al., 2004) to measure elements known to be important in text process and understanding including syntactic complexity, decoding, lexical sophistication, morphological complexity, cohesion, sentiment, and semantics (Collins-Thompson, 2014). The features were used in a principal components analysis to extract components of reading. Each tool and the features it measures are discussed briefly below (as well as in Table 1). The tools calculate features based on tokens and word types as well as features for all words, content words, and function words. The features are also normed by text length to provide mean and standard deviation scores. In total, 813 SALAT variables were initially selected for consideration.

Sentiment analysis and cognition engine (SEANCE)

SEANCE 1.2 is a sentiment analysis tool that measures features related to emotions, social positioning, and cognition (Crossley, Kyle, et al., 2017). Positive emotional features in texts have shown meaningful relationships with text comprehension and can indicate pragmatic choices that influence text readability (Collins-Thompson, 2014). Features related to social positioning and cognition have also shown relationships with readability. For instance, cognitive terms that capture larger semantic units like temporality, spatiality, states, and goals have shown meaningful relationships with text

Table 1. NLP tools used in this analysis.

Tool	Features	Website with additional information
TERA	Calculates component scores of reading related to narrativity, syntax, concreteness, and cohesion based on linguistic features found in Coh-Metrix.	https://soletlab.adaptiveliteracy.com:8443/ http://cohmetrix.memphis.edu/cohmetrixhome/
SEANCE	Reports on thousands of lexical categories and 20 component scores related to sentiment, social cognition, and social order.	https://www.linguisticanalysistools.org/seance.html
TAACO	Calculates 150 indices of both local and global cohesion, including a number of type–token ratio indices (including specific parts of speech, lemmas, bigrams, trigrams and more), adjacent overlap indices (at both the sentence and paragraph level), and connectives indices	https://www.linguisticanalysistools.org/taaco.html
TAALED	Calculate a wide variety of lexical diversity indices related to volume, abundance, and variety.	https://www.linguisticanalysistools.org/taaled.html
TAALES	Measures over 400 classic and newly developed indices of lexical sophistication including a wide range of sub-constructs related to word recognition, word neighbors, contextual distinctiveness, word frequency, and psycholinguistic norms.	https://www.linguisticanalysistools.org/taales.html
TAMMI	Calculates measures related to basic morpheme counts, morphological variety, morphological complexity, morpheme type–token counts, and variables found in the MorphoLex database	https://www.linguisticanalysistools.org/tammi.html
TAASSC	Measures a number of indices related to syntactic development including classic indices of syntactic complexity (e.g., mean length of T-unit) and fine-grained indices of phrasal (e.g., number of adjectives per noun phrase) and clausal (e.g., number of adverbials per clause) complexity.	https://www.linguisticanalysistools.org/taassc.html

comprehension (Brezina & Pallotti, 2019). For sentiment analysis, SEANCE pull information from a number of word vectors including the Affective Norms for English Words (ANEW) database (M. M. Bradley & Lang, 1999), which includes affective norms for valence, pleasure, arousal, and dominance (Osgood et al., 1957). SEANCE also calculates valance shifting by ignoring positive terms that are negated and uses a part-of-speech (POS) tagger to calculate counts specific to nouns, verbs, and adjectives. For social positioning and cognition categories, SEANCE calculates features based on the General Inquirer (GI; Stone et al., 1966), which includes 119 word-lists organized into 17 semantic categories. These categories cover affective, social, political, and psychological constructs including pleasure, overstatements, institutions, roles, social categories, references to places, references to objects, communication, motivation, cognition, and assent.

Tool for the Automatic Analysis of Cohesion (TAACO)

TAACO 2.0.4 (Crossley et al., 2016; Crossley, Skalicky, et al., 2019) derives measures of text cohesion that examine connectivity between segments of text (Halliday & Hasan, 1976) including features of local and global cohesion, both of which are important components of discourse processing (McNamara & Magliano, 2009). TAACO features include lexical and semantic overlap for words, content words, and function words across adjacent sentences (local cohesion) and adjacent paragraphs (global cohesion). Overlap indices are calculated as a percentage of overlap or as binary operators. TAACO also calculates incidence of connective in a text including logical connectives (i.e., words that connect sentences uses words like moreover and nevertheless) and temporal connectives (i.e., words that connect sentence in time using words like after and before).

Tool for the automatic analysis of lexical diversity (TAALED)

TAALED 1.4.1 measures features related to three dimensions of lexical diversity: volume (number of tokens), abundance (number of types), and variety (the relative proportion of unique words; Kyle et al., 2020). Lexical diversity features capture the range and diversity of vocabulary in text and assess the likelihood of a text incorporating a larger number of unique words and/or greater variation in vocabulary, both of which lead to greater text processing difficulty (Collins-Thompson, 2014). Indices are also calculated using all lemmas, content lemmas, and function lemmas (Kyle et al., 2020). For variety, TAALED calculates traditional type–token ratio features (TTR) and a TTR indices designed to control for text length confounds (Jarvis, 2013). These features include mean segmental type–token ratio (MSTTR), which measures the average type–token ratio over subsamples of a given number of words in a text (Johnson, 1944) and the Measure of Textual Lexical Diversity (MTLD), which calculates the mean length of sequential tokens that fall above a given value (McCarthy & Jarvis, 2010).

Tool for the automatic analysis of lexical sophistication (TAALES)

TAALES 2.2 (Kyle & Crossley, 2015; Kyle et al., 2018) reports on features related to decoding and lexical production at the word and phrasal level. Lexical features of an important component of text readability because the inclusion of words that are difficult or unfamiliar to readers will decrease text comprehension (Collins-Thompson, 2014). TAALES reports on lexical properties including word familiarity, word imageability using the MRC database (Coltheart, 1981), concreteness using the norms reported in Brysbaert et al. (2014), and age of acquisition scores found in Kuperman et al. (2012). TAALES includes measures related to the number of associations incorporated into words or phrases including semantic distinctiveness calculations (SEM-D, Hoffman et al., 2013), word meaningfulness scores reported in the MRC (Coltheart, 1981), the number of word associations as found in the Edinburgh Association Thesaurus (Kiss et al., 1973), semantic similarity scores derived from Latent Semantic Analysis (LSA; Landauer et al., 1998), and McDonald co-occurrence probability (McDonald & Shillcock, 2001). TAALES also provides measurements of word frequency calculated using the SUBTLEXus corpus (Brysbaert & New, 2009), the Hyperspace Analogue to Language (HAL) corpus (Lund & Burgess, 1996), and the Contemporary Corpus of American English (COCA; Davies, 2010). In addition, TAALES reports on a variety of features found in the English Lexicon Project

database (Balota et al., 2007) including word neighborhood effects, lexical decision times, and word name latencies and accuracies.

Tool for automatic measurement of morphological information (TAMMI)

TAMMI 2.0 (Crossley et al., 2024) includes measures related to basic morpheme counts (e.g., number of suffixes, number of morphemes, and suffix length), morphological variety, morphological complexity, morpheme type–token counts, and variables found in the MorphoLex database (Sánchez-Gutiérrez et al., 2018) including morpheme frequency/length, morpheme family size counts and frequency, and morpheme hapax counts. TAMMI also calculates morphological complexity indices (MCI) based on Brezina and Pallotti (2019). In addition, TAMMI calculates morphological complexity indices based on morphological variety and type–token ratios for both inflectional and derivational morphemes. Decoding words is primary component of reading and words that contain many morphemes or infrequent morphemes may require readers to not just parse the root meaning of the word but also the inflectional and derivational morphemes attached to it. A higher incidence of morphologically complex words in a text will be harder to decode and thus more difficult to read (Berninger et al., 2003, Nagy & Anderson, 1984).

Tool for the Automatic Analysis of Syntactic Sophistication and complexity (TAASSC)

TAASSC 1.3.8 (Kyle, 2016; Kyle & Crossley, 2017) measures clausal and phrasal indices of syntactic complexity and indices related to complexity of verb-argument constructions (defined as a main verb plus all of its direct dependents). More complex syntactic features have been shown to slow text processing, and measures of syntax taken from deep parses of sentence structures can reliably measure text readability (Collins-Thompson, 2014; Crossley, Kyle, et al., 2019). Measures of clausal complexity include passive agents, prepositions, or nominal per clause. Measures of phrasal complexity include determiners, possessives, prepositions, or dependents per nominal, object, or direct object. A verb argument construction consists of a verb slot and the related arguments in a syntactic construction (Kyle & Crossley, 2017). For instance, the verb *give* in English requires three arguments (agent, theme, and recipient) as seen in the sentence *She* (agent) *gave the paper* (theme) *to the boy* (recipient). Verb argument construction features are measured using the COCA corpus and include the count of common constructions, the frequency of constructions, construction variety (measured as a type/token ratio), and collostruction ratios (i.e., the ratios of words that are attracted or repulsed to constructions).

Statistical analysis

All statistical analyses were conducted in R 3.6.1 (R Core Team, 2021) to enhance replication. To develop linguistic components of text readability, we used a Principal Components Analysis (PCA). To assess the strength of the derived components to predict the reading ease scores, along with those reported by Graesser et al. (2011), we used correlations and linear multiple regressions. To assess similarities between the newly developed components and the Graesser et al. components, we used correlations. We presumed linearity in the correlational analyses because of theoretical links between the linguistic features selected and text readability.

To examine the potential for underlying constructs related to readability in the CLEAR corpus, and to reduce the linguistic features into related sets of measures, a principal component analysis (PCA) was conducted using the psych package (Revelle, 2023). We selected a PCA over an exploratory factor analysis (EFA) because our interest was in reducing the data to a manageable number of components that reflected composite scores for the observed variable in a similar fashion to Graesser et al. (2011).

Prior to the PCA, the linguistic features reported by SALAT were checked to ensure normal distribution of variables. First, any variables that reported zero counts for more than 20% of the essays were removed. This process removed 256 variables. Next, variables that reported skewness and kurtosis greater than 3 were removed resulting in additional 134 variables being removed. Variable

transformations were not attempted because many variables reported zero values (making logarithmic transformation impossible) and/or negative values (making square root transformation impossible). We also checked for multicollinearity between the variables through correlational analyses using the cor.test() function (R core team, 2021). If two variables reported a correlation > .899, the variable that reported a lower correlation with the reading ease scores was removed. This was done to remove linguistic variables that were redundant (e.g., TAALES reports over 50 word frequency metrics). After data pruning, we were left with 169 linguistic features for our PCA analysis (i.e., 254 additional variables were removed because of multi-collinearity). These variables were standardized and included in a PCA with an oblique rotation (promax). We used a promax rotation because we presumed the resulting components would be correlated. This is in contrast to Graesser et al. (2011), who used a varimax rotation, which presumes that the resulting components would be orthogonal. We selected an oblique rotation based on recommendations from Biber (1988) who argued that linguistic features are correlated by their very nature, Sheehan et al. (2015), who used an oblique rotation when developing the components for the TextEvaluator tool, and from Costello and Osborne (2005) who argued that the default expectation in the social sciences is some correlation among components. Our thinking was also influenced by Goulart and Wood (2019), who conducted a meta-analysis of dimension reduction research in linguistics and found that 83% of researchers used an oblique versus and orthogonal rotation. We only kept variables in the PCA if they showed an absolute component loading of at least .30 with a component (Biber, 1988). Once we had a final solution from the oblique rotation, we calculated weighted factor scores for each component derived from the PCA.

To examine associations between the derived SALAT components and the reading ease scores, we used Pearson correlations, calculated using the cor.test() function in R. We then used a ten-fold cross-validated stepwise linear model to predict the reading ease scores using the derived component scores. We used a stepwise model because we did not have a-priori support for ordering the entry of variables. For the stepwise regression, we used the leapSeq function in Leaps and Caret Kuhn (2008). We also ran simple linear regression models using the lm() function (R core team, 2021) so that we could compute the relative importance of the indices in each model using the calc.relimp() function in the relaimpo package (Grömping, 2006). Specifically, the metric lmg (Lindeman et al., 1980), which takes into account both the direct relationship between the independent and dependent variables (i.e., the bivariate correlation) and the indirect relationship between the independent and dependent variables (i.e., the amount of variance explained when included in a multivariate model) was used. A similar process was used for the TERA components. To assess differences in prediction strengths for the reading ease scores between the SALAT components and the TERA components, a Fisher's r-to-z transformation was conducted. A correlational analysis was conducted to assess similarities between the SALAT and TERA components.

RESULTS

Principle component analysis (SALAT features)

A Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy indicated that two variables needed to be removed (i.e., their KMO value were below .5, Kaiser & Rice, 1974). Once these two variables (incidence of inflected tokens and number of roots per content word, both calculated from TAMMI) were removed, all individual variables reported KMO values above 5 and the overall KMO score = .906 indicated a meritorious sample. The PCA reported a Bartlett's test of sphericity, $\chi 2$ (4724) = 907,081.9, $p < .001$, indicating that correlations between the linguistic features were sufficiently large for the PCA.

An initial PCA with all possible factors ($n = 167$) indicated 34 components that reported eigen values over 1. Within the components, there was a slight break in the cumulative variance explained between the seventh and eighth component (see the Scree Plot in Figure 1). Considering this break, we decided on a seven-component solution and a PCA was conducted extracting only seven components. An initial PCA indicated that 27 variables did not load at or above the .30 threshold. These variables were removed and a final seven component PCA with the 140 variables was conducted. Descriptive statistics for this PCA are

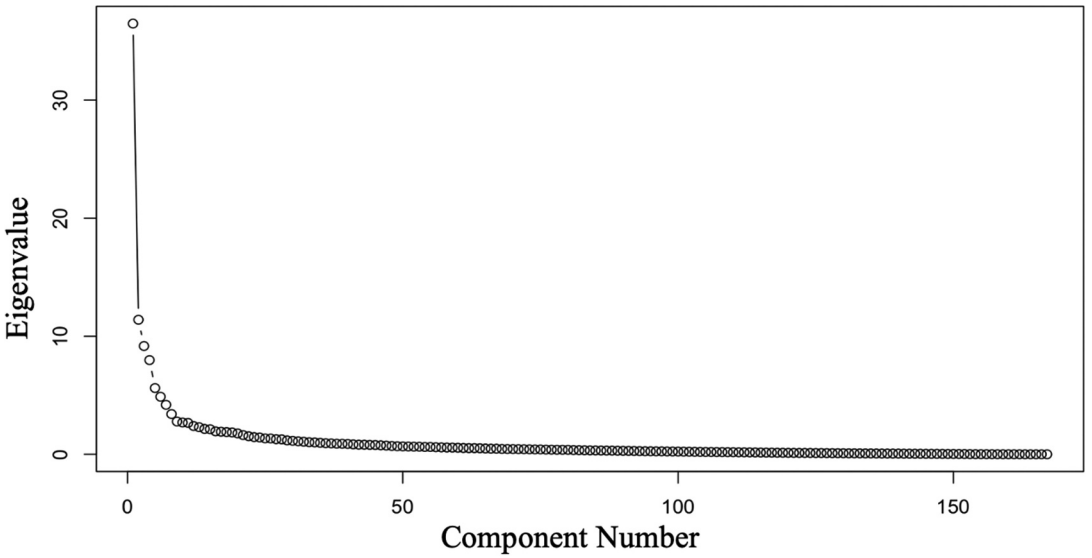


Figure 1. Scree plot from component analysis.

reported in Table 2. None of the components were strongly correlated with one another, although small-to-medium correlations are reported. This is expected within language studies where the default presumption is correlation among linguistic components (Biber, 1988; Costello & Osborne, 2005). All components demonstrated at least a weak correlation ($r > .099$) with the reading ease scores (see Table 3). We discuss each of the individual components, their weights, and their interpretation below.

Component 1 (informational)

The first component consisted of 47 features and explained 18% of the variance in the PCA. The component reported an r of −.569 with the reading ease scores indicating that the component led to excerpts that were more difficult to read. The feature in the component with positive loadings, which are scaled eigenvectors, were generally related to lexical and morphological elements that made processing more difficult. There were, in addition, a number of semantic features related to aspects of narrative (e.g., time, space, description, and knowledge) that were negatively weighted in the component. Additionally, the component included a clausal syntactic feature that was negatively weighted and a phrasal feature that was positively correlated. The component was labeled "Informational" because of the difficulty of the words, the negative loadings of features related to narrativity, and the phrasal density feature. The features included, their weights, and their explanation are reported in Table 4.

Component 2 (common syntactic constructions)

The second component comprised 12 features and explained 8% of the variance in the PCA. The component reported an r of .391 with the reading ease scores, indicating that the component led to excerpts that were easier to read. The features that positively informed this component were generally related to the proportion of verb-argument constructions found in the excerpts that were common in large-scale corpora. The component also included indices that measure the construction frequency. In short, excerpts that were easier to read included a larger proportion of frequent verb-argument construction. Negative weights for this component were construction variety and features related to clausal complexity, indicating that a lower variety of constructions and less complex clauses lead to increased readability. This component was labeled common syntactic constructions because of its focus on verb argument constructions and

Table 2. Descriptive statistics for PCA.

Component	Loadings	Proportion variance	Cumulative variance
Comp 1: Informational	24.94	0.18	0.18
Comp 3: Noun phrase complexity	12.79	0.09	0.27
Comp 4: Lexical ease	11.15	0.08	0.35
Comp 2: Common syntactic structures	9.14	0.07	0.41
Comp 5: Function word ease	5.81	0.04	0.46
Comp 6: Narrativity	5.54	0.04	0.50
Comp 7: Lexical variety	5.75	0.04	0.54

Table 3. Correlations between reading ease scores and derived components.

Feature	Comp 1	Comp 3	Comp 4	Comp 2	Comp 5	Comp 6	Comp 7
Text readability	−0.569	−0.339	0.157	0.391	−0.213	0.309	−0.262
Comp 1: Informational	1	0.434	−0.243	−0.268	0.156	−0.409	0.055
Comp 3: Noun phrase complexity		1	−0.407	−0.272	0.232	−0.300	−0.007
Comp 4: Lexical ease			1	0.220	−0.101	0.230	−0.102
Comp 2: Common syntactic structures				1	−0.055	0.098	−0.274
Comp 5: Function word ease					1	−0.156	0.006
Comp 6: Narrativity						1	−0.018
Comp 7: Lexical variety							1

NP = Noun phrase.

clausal complexity. The features included, their weights, and their explanation are reported in Table 5.

Component 3 (noun phrase complexity)

The third component included 28 features and explained 9% of the variance in the PCA. The component reported an *r* of −.339 with the reading ease scores indicating that the component led to excerpts that were harder to read. Features that loaded positively on the component were related to the number of determiners per text and noun phrase, the number of dependents per noun phrases, and lexical indices that indicated easier to process function words (e.g., more frequent and familiar function words that had more neighbors). Features that loaded negatively were related to more possessive per noun phrase and lexical features that were reverse scaled. This component was labeled noun phrase complexity because it highlights elements of noun phrases including determiners, function words, nominals, and possessives. The features included, their weights, and their explanation are reported in Table 6.

Component 4 (lexical ease)

The fourth component included 20 features and explained 8% of the variance in the PCA. The component reported an *r* of .157 with the reading ease scores indicating that the component led to excerpts that were easier to read. Positive loadings were reported for semantic distinctiveness, word frequency, word neighbors, and word familiarity. Negative loadings were reported for word concreteness, imageability, and similarity, co-occurrence probability. This component was labeled lexical ease because the features related to lexical features that make processing easier. The features included, their weights, and their explanation are reported in Table 7.

Component 5 (function word ease)

The fifth component consisted of 10 features and explained 7% of the variance in the PCA. The component reported an *r* of −.213 with the reading ease scores indicating that the component led to excerpts that were harder to read. The loadings indicated that function words that were easier to process (e.g., more frequent neighbors, more neighbors, greater range, stronger co-occurrence

Table 4. Component 1: informational ($r = -.569$ with text readability).

Feature	Loading	Interpretation
Orthographic neighbors (Levenshtein Distance)	0.92	More lexical complexity
Number of suffixes (CW)	0.92	More morphological complexity
Word age of acquisition (Kuperman)	0.90	More lexical complexity
Number morphemes (CW)	0.90	More morphological complexity
Word age of exposure (average)	0.85	More lexical complexity
Suffix family size (CW)	0.83	More morphological complexity
Prefix length (CW)	0.82	More morphological complexity
Prefix frequency (CW)	0.82	More morphological complexity
Prefix family size (CW)	0.82	More morphological complexity
Orthographic neighbors (HAL frequency)	−0.81	More lexical complexity
Orthographic Neighbors (CW)	−0.79	More lexical complexity
Orthographic neighbors, (HAL frequency, CW)	−0.79	More lexical complexity
Word type associations (EAT CW)	0.76	More lexical complexity
Word age of exposure (slope)	0.72	More lexical complexity
Word imageability (MRC)	−0.72	More lexical complexity
Morpheme complexity (derivational)	0.72	More morphological complexity
Derivational morpheme TTR	0.70	More morphological complexity
Word naming time	0.69	More lexical complexity
Lexical decision time	0.67	More lexical complexity
Word naming time (SD CW)	0.65	More lexical complexity
Word range (SUBTLEXus, AW)	−0.62	More lexical complexity
Phonological neighbors	−0.62	More lexical complexity
Word associations (USF, CW)	−0.62	More lexical complexity
Word concreteness (Brysbaert)	−0.61	More lexical complexity
Word associations (EAT AW)	0.60	More lexical complexity
Word meaningfulness (MRC)	−0.60	More lexical complexity
Character bigram frequency (CW)	0.58	More lexical complexity
Word range (SUBTLEXus, CW)	−0.57	More lexical complexity
Morpheme family frequency (suffix, CW)	0.56	More morphological complexity
Morpheme family frequency (prefix, CW)	0.53	More morphological complexity
Word frequency (COCA fiction)	−0.53	More lexical complexity
Word naming (SD)	0.51	More lexical complexity
Lexical decision time (SD, CW)	0.50	More lexical complexity
Number of subjects per clause	−0.49	Less clausal complexity
Descriptive words (GI)	−0.48	Fewer descriptive words
Words related to knowledge (GI)	0.48	More words indicating knowledge
Travel words (GI)	−0.46	Fewer words related to travel
Word valence (ANEW)	−0.45	Fewer negative words
Abstract words (GI)	0.44	Fewer abstract words
Time and space words (Lasswell)	−0.43	Fewer time and space words
Character bigram frequency	0.40	More lexical complexity
Suffix length (CW)	0.40	More morphological complexity
Prepositions per nominal	0.39	More noun phrase complexity
Verb construction faith (COCA magazine)	0.34	More expected verb constructions
Word token associations (EAT)	0.34	More lexical complexity
Words related to time (GI)	−0.33	Fewer time words
Word token associations (EAT, FW)	0.32	More lexical complexity

SD = standard deviation, CW = content words, FW = function words, HAL = hyperspace Analogue to Language, GI = General Inquiry, ANEW = Affective Norms for English Words, EAT = Edinburgh Association Thesaurus, COCA = Corpus of Contemporary American English, SUBTLEXus = Subtitle frequency, USF = University of South Florida Norms, MRC = Medical Research Council.

probability, and lower ager of acquisition scores) informed this component. As a result, this component was labeled function word ease because the variables related to lexical features in function words that make processing easier. The features included, their weights, and their explanation are reported in Table 8.

Component 6 (narrativity)

The sixth component comprised 12 features and explained 4% of the variance in the PCA. The component reported an r of .309 with the reading ease scores indicating that the component led to

Table 5. Component 2: common syntactic constructions ($r = .391$ with text readability).

Feature	Loading	Interpretation
Common constructions (COCA academic)	0.87	Greater percentage of common lemma constructions
Common constructions (COCA magazine, lemma)	0.86	Greater percentage of common lemma constructions
Common constructions (COCA, lemma)	0.86	Greater percentage of common lemma constructions
Common constructions (COCA academic, lemma)	0.84	Greater percentage of common lemma constructions
Common constructions (COCA fiction)	0.84	Greater percentage of common lemma constructions
Common constructions (COCA fiction, lemma)	0.81	Greater percentage of common lemma constructions
Common constructions (COCA magazine)	0.79	Greater percentage of common lemma constructions
Construction frequency (COCA)	0.71	More frequent constructions
Preposition per clause	−0.63	Fewer prepositions per clause
Construction frequency (COCA fiction)	0.61	More frequent constructions
TTR constructions (COCA fiction)	−0.59	Lower construction TTR
Dependents per clause (SD)	−0.53	Less variation in dependents per clause

SD = standard deviation, COCA = Corpus of Contemporary American English, TTR = Type Token Ratio.

Table 6. Component 3: noun phrase complexity ($r = -.339$ with text readability).

Feature	Loading	Interpretation
Determiners per nominal	1	More noun phrase complexity
Word frequency (HAL)	0.99	More word complexity
Number of determiners	0.97	More noun phrase complexity
Word frequency (HAL, FW)	0.94	More function word complexity
Word frequency logarithm (HAL, FW)	0.75	More function word complexity
Determiners per nominal subject	0.75	More noun phrase complexity
Phonological neighbors (Levenshtein distance, FW)	0.72	More function word complexity
determiners per nominal subject (no pronouns)	0.71	More noun phrase complexity
Word type associations (EAT)	0.68	More function word complexity
Word meaningfulness (MRC, FW)	−0.66	More function word complexity
Phonological Neighborhood Frequency (FW)	0.60	More function word complexity
Phonographic neighbors (FW)	−0.60	More function word complexity
Phonological Neighbors (FW)	−0.56	More function word complexity
Word naming scores (z value, FW)	0.55	More function word complexity
Phonological neighbors (HAL frequency)	0.53	More function word complexity
Determiners per direct object	0.51	More noun phrase complexity
Dependents per nominal subject	0.51	More noun phrase complexity
Dependents per nominal (no pronouns)	0.48	More noun phrase complexity
Semantic Distinctiveness (SEM, FW)	0.47	More function word complexity
Word familiarity (MRC, FW)	0.46	More function word complexity
Dependents per nominal	0.46	More noun phrase complexity
Dependents per nominal subject (no pronouns)	0.45	More noun phrase complexity
Lexical decision time (SD, FW)	0.44	More function word complexity
Possessives per nominal (no pronouns)	−0.41	More possessives per nominal
Dependents per nominal subject (SD)	0.33	More noun phrase complexity
Possessives per object of the preposition	−0.32	More noun phrase complexity
Dependents per object of the preposition (no pronouns)	0.31	More noun phrase complexity
Dependents per object of the preposition	0.30	More noun phrase complexity

SD = standard deviation, CW = content words, FW = function words, HAL = hyperspace Analogue to Language, EAT = Edinburgh Association Thesaurus, MRC = Medical Research Council, SEM = Semantic Distinctiveness.

excerpts that were easier to read. Positive loadings were reported for words related to action and activities, social relations, hostility, carrying, completion of tasks, goals, work, change, and submission). Negative loadings were reported from construction frequency. This component was labeled narrativity because the variables focus on narrative semantics and less frequent constructions found in written language. The features included, their weights, and their explanation are reported in Table 9.

Component 7 (lexical variety)

The seventh component included 11 features and explained 4% of the variance in the PCA. The component reported an r of −.262 with the reading ease scores, indicating that the component led to

Table 7. Component 4: lexical ease ($r = .157$ with text readability).

Feature	Loading	Interpretation
Semantic distinctiveness (SEM-D)	0.88	More contextually variable
Word frequency (logarithm, HAL, AW)	0.87	More frequent words
Word imageability (MRC, CW)	−0.74	More imageable words
Word frequency (logarithm, HAL, CW)	0.73	More frequent words
Word concreteness (Brysbaert, CW)	−0.7	Less concrete words
Word frequency (SUBTLEXus, CW)	0.68	More frequent words
State words (GI)	0.59	More state verbs
Word similarity (LSA top cosine)	−0.58	Less similar words
Orthographic Neighborhood Frequency (CW)	0.56	More frequent neighbors
Lexical type density	−0.52	Less lexical density
Phonological Neighborhood Frequency (CW)	0.52	More word neighbors
Co-occurrence probability (McDonald, CW)	−0.48	Less co-occurrence probability
Word familiarity (MRC, AW)	0.48	More familiar words
Word familiarity (MRC, CW)	0.44	More familiar words
Word co-occurrence probability (SEM)	−0.42	Less co-occurrence probability
Number of function word types	0.42	More function word variety
Root morpheme frequency	0.42	More freq word
Words related to solving problems (GI)	0.38	More words related to solving problems
Words related to need (GI)	0.36	More words related to the expression of need or intent
Root morpheme frequency (logarithm)	0.35	More frequent words

CW = content words, FW = function words, HAL = hyperspace Analogue to Language, MRC = Medical Research Council, GI = General Inquiry, SUBTLEXus = Subtitle frequency, LSA = Latent Semantic Analysis, SEM = Semantic Distinctiveness.

Table 8. Component 5: function word ease ($r = -.213$ with text readability).

Feature	Loading	Interpretation
Orthographic neighborhood frequency (FW)	0.83	More frequent neighbors
Orthographic neighborhood frequency (HAL, FW)	0.80	More frequent neighbors
Phonographic neighbors (FW)	0.79	More frequent neighbors
Orthographic neighborhood frequency	0.68	More frequent neighbors
Phonographic neighbors	0.65	More frequent neighbors
Age of acquisition (Kuperman, FW)	−0.59	Lower age of acquisition
Co-occurrence probability (McDonald, FW)	0.47	Higher co-occurrence probability
Lexical decision accuracy (FW)	−0.46	Lower lexical decision accuracy
Word range (SUBTLEXus, FW)	0.38	Words with greater range
Lexical decision accuracy	−0.36	Lower lexical decision accuracy

FW = function words, HAL = hyperspace Analogue to Language, SUBTLEXus = Subtitle frequency.

Table 9. Component 6: narrativity ($r = .309$ with text readability).

Feature	Loading	Interpretation
Action words	0.78	More action words
Active words	0.73	More words implying an active orientation
Common constructions (COCA fiction, lemma)	−0.70	Lower percentage of common lemma constructions
Construction frequency (COCA news)	−0.68	Less frequent constructions
Construction frequency (COCA)	−0.66	Less frequent constructions
Social words	0.52	More words with social implications
Hostile words	0.42	More words indicating hostility
Carrying words	0.38	More words related to carrying objects
Work words	0.36	More words related to work
Completion words	0.35	More words indicating goal achievement
Goal words	0.34	More words indicating goal driven activities
Change words	0.30	More words indicating change

GI = General Inquiry, COCA = Corpus of Contemporary American English.

excerpts that were harder to read. The results indicated that lexical variation measures related to TTR and abundance positively loaded on the component. Features related to lexical overlap between sentences loaded negatively.

This component was labeled lexical variety because greater variety of words led to higher reading ease scores while repetition of words across sentences led to lower reading ease scores. The features included, their weights, and their explanation are reported in Table 10.

Regression Model (SALAT components)

The seven SALAT components were entered into a linear model to predict the reading ease scores. An initial ten-fold cross validation model using step-wise selection showed a suppression effect for the fourth component (lexical ease) and it was removed from the analysis. A second regression model reported an $r = .662$, $R^2 = .438$ and retained all six of the remaining components. A follow-up linear regression without cross-validation using these six variables reported $F(6, 4717) = 610.9$, $p < .001$. The relative importance metrics indicate that the strongest components of reading ease was the informational component followed by common syntactic constructions, narrativity, noun phrase complexity, lexical variety, and function word ease (see model parameters summarized in Table 11). The strongest predictor was the Informational Component followed by the Common Syntactic Structures, Lexical Variety, Noun Phrase Complexity, Narrativity, and Function Word Ease Components. Post-hoc tests including variance inflation factors (for multi-collinearity), Q–Q plots (for assessing the normality of the residuals), and a scatterplot of residuals versus fitted predictions (for homoscedasticity) indicated that all linear model assumptions were met.

Regression Model (TERA components)

Correlations

Of the eight TERA components assessed, none were removed because of multi-collinearity. However, four were removed from the regression analyses because they did not report at least a weak correlation ($r < .099$) with the reading ease scores. The components removed were related to deep cohesion, verb overlap, connectives, and temporal cohesion. Correlations among the remaining four variables indicated small-to-medium relationships ($r < .10$ and $> .50$). Correlations for the TERA components are reported in Table 12.

Linear Model

The remaining four components from TERA were entered into a linear model to predict the reading ease scores. An initial ten-fold cross validation model using step-wise selection reported an $r = .596$, $R^2 = .355$ and retained all four of the components. A follow-up linear regression without cross-validation using these four variables reported $F(4, 4719) = 648.6$, $p < .001$. The relative importance metrics indicate that the strongest components of reading ease were the narrativity component

Table 10. Component 7: lexical variety ($r = -.262$ with text readability).

Feature	Loading	Interpretation
Mean segmental TTR (50 words, CW)	0.79	More lexical variety
MTLD (moving average wrap, CW)	0.75	More lexical variety
TTR (AW)	0.69	More lexical variety
Sentence word overlap (binary, CW)	−0.58	Less overlap of words between sentences
Sentence word overlap	−0.56	Less overlap of words between sentences
Number of word types (CW)	0.56	More lexical variety
TTR (COCA fiction, lemma)	0.45	More lexical variety
Sentence verb overlap (binary)	−0.37	Less overlap of words between sentences
Words indicating passivity	0.36	More passive words
Words implying weakness	0.35	More weakness words
Lexical decision time (SD)	0.34	Greater lexical decision time variation

SD = standard deviation, CW = content words, GI = General Inquiry.

Table 11. Linear model to predict reading ease scores using SALAT components.

Variable	Relative Importance	Estimate	SE	t	p
(Intercept)		−0.958	0.011	−84.831	<.001
Comp 1: Informational	0.479	−0.452	0.014	−33.573	<.001
Comp 2: Common syntactic structures	0.180	0.204	0.012	16.379	<.001
Comp 7: Lexical Variety	0.107	−0.189	0.012	−16.001	<.001
Comp 3: Noun phrase complexity	0.095	−0.049	0.013	−3.754	<.001
Comp 6: Narrativity	0.088	0.079	0.013	6.280	<.001
Comp 5: Function word ease	0.051	−0.114	0.012	−9.768	<.001

Table 12. Correlations between TERA components and reading ease scores.

Variable	Comp 1	Comp 2	Comp 3	Comp 4	Comp 5	Comp 6	Comp 7	Comp 8
Reading easy score	0.387**	0.313**	0.255**	0.15**	0.048**	0.02	−0.063**	0.041**
Component 1: Narrativity	1	−0.082**	0.101**	0.184**	0.215**	−0.1**	−0.084**	0.023
Component 2: Syntactic simplicity		1	−0.15**	−0.179**	−0.038**	−0.192**	0.029*	−0.051**
Component 3: Word concreteness			1	0.129**	−0.036**	−0.094**	−0.242**	0.07**
Component 4: Referential cohesion				1	0.081**	−0.114**	0.003	0.014
Component 5: Deep cohesion					1	−0.013	−0.194**	0.084**
Component 6: Verb overlap						1	0.003	0.064**
Component 7: Connectives							1	−0.132
Component 8: Temporal cohesion								1

** $p < .010$, * $p < .050$.

followed by syntactic simplicity, word concreteness, and referential cohesion (see model parameters summarized in Table 13). Post-hoc tests including variance inflation factors, Q–Q plots, and a scatterplot of residuals versus fitted predictions indicated that all linear model assumptions were met.

Comparison between SALAT and TERA models

We ran a simple Fisher r-to-z transformation to assess the significance of the differences between the *r* values reported in the TERA component regression model and the SALAT component regression model. The Fisher's test reported significant differences between the two models ($z = 5.49$, $p < .001$), although this *p* value is likely inflated because of sample size. These results follow the patterns in the regression analyses, which indicate that the SALAT model explained ~8% more variance in the readability scores than the TERA models.

Correlation comparison among components

We conducted a correlational analysis between the TERA components reported in Graesser et al. (2011) and the new SALAT components developed in this paper (see Table 14). The results showed that the TERA narrativity component was strongly correlated with the SALAT informational component (negative), noun phrase complexity (negative), lexical ease (positive) and narrativity (positive). The TERA syntactic simplicity component was also moderately correlated with the SALAT common syntactic constructions

Table 13. Linear model to predict reading ease scores using TERA components.

Variable	Relative Importance	Estimate	SE	t	p
(Intercept)		−1.006	0.016	−62.170	<.001
Comp 1: Narrative	0.400	0.374	0.012	31.105	<.001
Comp 2: Syntactic similarity	0.363	0.342	0.010	33.665	<.001
Comp 3: Word concreteness	0.185	0.232	0.011	22.006	<.001
Comp 4: Referential cohesion	0.053	0.109	0.011	9.874	<.001

Table 14. Correlations between TERA and SALAT components.

Components	SALAT: Informational	SALAT: Common syntactic constructions	SALAT: Noun phrase complexity	SALAT: Lexical ease	SALAT: Function word ease	SALAT: Narrativity	SALAT: Lexical variety
TERA: Narrativity	−0.705**	0.170**	−0.667**	0.583**	−0.103**	0.562**	0.100**
TERA: Syntactic simplicity	−0.212**	0.467**	−0.195**	−0.052**	−0.067**	0.246**	−0.352**
TERA: Word concreteness	−0.435**	−0.162**	0.055**	−0.436**	−0.130**	0.215**	0.105**
TERA: Referential Cohesion	−0.047**	0.105**	−0.007	0.120**	0.024	0.086**	−0.580**
TERA: Deep cohesion	−0.024	−0.115**	−0.241**	0.297**	−0.076**	0.187**	−0.089**
TERA: Verb overlap	−0.105**	0.075**	0.163**	0.335**	−0.015	−0.195**	0.005
TERA: Connectives	0.104**	0.163**	0.168**	−0.004	0.121**	0.068**	0.018
TERA: Temporal cohesion	−0.101**	0.004	0.018	−0.191**	0.016	−0.035*	0.003

** $p < .010$, * $p < .050$.

component (positive) and the lexical variety component (negative). The TERA component word concreteness was moderately correlated with the SALAT information component (negative) and lexical ease (positive). The TERA component referential cohesion was strongly correlated with the SALAT component lexical variety. No other moderate or strong correlations were reported between the other components.

DISCUSSION

The main goal of this study was to examine the potential to develop more robust language components related to text readability using theoretically inspired features derived from NLP tools. Language components as compared to individual linguistic features allow for related features to merge into larger constructs that are easier to interpret and provide greater applications in theory and practice. The study is a partial replication of Graesser et al. (2011), who developed eight components of reading using language features from the Coh-Metrix tool (i.e., the TERA components). The Graesser et al. study had a few limitations including potentially developing components using dependent text excerpts and the use of a corpus that did not allow for the assessment of relationships between the developed components and text readability. The latter concern was partially addressed in Graesser et al. (2014), where the TERA components were correlated with traditional readability formulas.

This study built on the original Graesser et al. study by using more advanced NLP tools reported by SALAT that reported on a greater breadth and depth of linguistic features including features related to sentiment, cognition, semantics, syntactic constructions and complexity, morphology, and lexical variety. Additionally, the components were developed on a corpus of excerpts (the CLEAR corpus) that were independent and were uniquely scored for readability. Statistical analyses indicated that seven components related to informational text, syntactic constructions, noun phrase complexity, lexical ease, function word ease, narrativity, and lexical variety predicted 44% of the variance in the reading ease scores reported in the CLEAR corpus, which is lower than the variance reported for a readability formula based on individual SALAT features (i.e., 51% of the variance reported in Crossley et al., 2023). The TERA components explained around ~36% of the variance in the reading ease scores. Strong to weak correlations were reported between the TERA components and the SALAT components. The results provide strong evidence that individual linguistic features related to text readability can be combined to form components that are strong predictors of readability.

Research question 1

Research question 1 related to extracting and interpreting the linguistic components derived from the reading excerpts using SALAT. An important question is whether the derived components overlap strongly with reading research and what they can tell us about language features that make a text easier or more difficult to read. A number of previous studies have indicated that syntactic complexity, lexical sophistication, and discourse are important components of text readability (Cunningham et al., 2005; Gernsbacher, 1990; Just & Carpenter, 1980; Mesmer et al., 2012) and this is reflected in five out of the seven components. For instance, research indicates that less complex syntactic structures lead to increased reading ease (Cunningham et al., 2005; Mesmer et al., 2012) and this is reflected in component 4 (common syntactic constructions) and component 2 (noun phrase complexity), both of which measure syntactic complexity to some degree. However, neither component aligns strongly with traditional notions of syntactic complexity, which is often defined as related to clausal complexity and sentence length (Klare, 1952). The features in these components also do not strongly align with the features in the TERA syntactic simplicity componet, which focus on sentence length, words before the main verb, and syntactic similarity. For example, SALAT component 4 encapsulates the use of common verb argument constructions and indicates that more common constructions lead to increased reading ease. SALAT component 2 captures the length of noun phrases and the incidences of noun phrase modifiers (i.e., noun phrase complexity) and indicates that more complex noun phrases lead to decreased reading ease. These two components provide evidence that features beyond clausal complexity, specifically, verb-argument constructions and noun phrase complexity, may be important indicators of text readability.

Research also indicates that lexical sophistication negatively influences text readability, such that more difficult lexical items lead to lower text readability (Howes & Solomon, 1951; Juel & Solso, 1981; Mesmer, 2005; Richardson, 1975). In terms of lexical sophistication, three SALAT components relied on lexical features including component 3 (lexical ease), component 6 (function word ease), and component 7 (lexical variety). Most work on lexical sophistication has focused on traditional lexical features like those found in component 3, which captures the difficulty of words in reference to semantic distinctiveness, word frequency, imageability, and concreteness. Like previous studies, component 3 indicates that texts with more difficult words are hard to read. Component 6 is unique because it focuses on function word features and indicates that more complex function words that are less frequent, have more fewer orthographic, phonetic, and phonographic neighbors, are acquired later, and are more difficult to process leading to decreased reading ease. Component 7 encapsulates lexical variety features based on type–token ratio counts and also includes measure of text cohesion related to overlap of words across sentences. Intuitively, this makes sense because type–token ratios and word overlaps counts both measure the repetition of tokens. Component 7 provides evidence that texts with fewer repeated words decrease text readability. Overall, the SALAT components indicate that readability may be related to the sophistication of words, the difficulty of function words and lexical repetition.

Components 1 and 5 best seem to capture language features common in informational and narrative discourse, respectively. This two component solution is different from the single TERA component that operationalized information and narrative within a continuum. The TERA component (narrativity) indicated that text difficulty was related to increased informational discourse and decreased narrativity. The SALAT informational component is similar to the TERA narrativity component because it includes features related both to informational and narrative text (although the strongest predictors are related to informational text). For instance, the component includes words and morphological features that were more difficult to process, a clausal syntactic feature common to spoken text that loaded negatively, and a noun phrase features common in written text (Biber et al., 2002). The component also featured indices common to narrativity that were negatively loaded including semantic features related to time, space, description, and knowledge. In contrast, the narrativity component included only features related to narrativity including semantic features related to action and activities, social relations, work, the completion of tasks, and the discussion of goals. The component also included more common verb argument constructions. Overall, the informational and narrative components indicated that texts that

include more aspects related to narrativity are easier to comprehend than texts comprising more features common in informational texts, which supports previous findings (Best et al., 2008; Sáenz & Fuchs, 2002). Unlike the Graesser et al. study, the Narrative Component was not the strongest predictor of text readability. It had the fourth strongest correlation with readability below that of the Informational, Noun Phrase Complexity, and Common Syntactic Structures Components. In the regression analysis, it was the least predictive component of text readability, explaining ~6% of the overall variance. In contrast, the Informational Component was the strongest predictor explaining 48% of the overall variance. This result likely hinges on the idea that the Informational Component contains some aspects of narrativity and that Narrative and Informational Component are negatively correlated with one another ($r = -0.409$), indicating some shared variance. This shared variance likely lowered the importance of the Narrative Component in the regression.

It should also be noted that few social or sentiment variables loaded into the seven components extracted from this analysis. A single variable related to emotional valence loaded into the informational component indicating that informational texts had fewer negative terms. Narrative texts included more social words and words related to hostility. Overall, the results indicate that cognition terms reported by SEANCE (e.g., words related to action and activities, completion, change, and goals, work, knowledge, abstraction, travel, space, and time) are important features of readability, but that the social and sentiment terms reported by SEANCE are not strong predictors of readability.

Research question 2

Research question 2 asked if the components derived from SALAT were similar to those reported in TERA. We find that there are a number of similarities between the reading components derived in this study and the TERA components reported in Graesser et al. (2011). Both studies yielded a component related to narrativity along with components associated with syntactic complexity and lexical sophistication. A difference between the two studies is that Graesser et al. reported five components related to cohesion including referential cohesion, deep cohesion, connectives, and temporal cohesion. This result makes sense because TERA components were developed using Coh-Metrix, which focused on cohesion. In contrast, no SALAT components were strongly or uniquely related to cohesion with only a small number of cohesion features that capture word overlap loading into the lexical variety component (component 7). It should be noted that the correlations between the SALAT lexical variety component and the TERA referential cohesion component correlated at ~.60, which is a strong correlation. This is likely because the TERA referential cohesion component's strongest loadings were for variables related to word overlap among sentences and the weakest loadings were related to type–token ratio. This is the opposite of the SALAT components where the strongest loadings were related to type–token ratio and the weakest related to word overlap, but the results do indicate that local cohesion features like word overlap are related to features that also measure repetition of words like type–token ratio counts.

Additionally, the current study yielded multiple components related to both syntactic complexity and lexical sophistication, whereas the Graesser et al. study reported only one for each construct. This is also likely the result of the number of features sampled. The Graesser et al. PCA included ~50 features (sampled from over 100) whereas the current PCA included 140 features (sampled from over 800). As noted earlier, many of the features in the Graesser study were also related to cohesion, which may be the result of not controlling for collinearity. In the current study, over 80% of the normally distributed cohesion variables from TAACO were removed because of either multi-collinearity of weak overlap with readability criteria.

Research question 3

Our third research question examined how well the new SALAT components perform compared to the TERA components. We find that the components developed in the current study outperform the TERA components in terms of explaining text readability. Individually, all the components from the current study showed at least a weak effect ($r > .099$) with measures of text readability, while four of the seven components showed medium to strong effects with text readability ($r > .299$). In contrast, four of the eight components reported in Graesser et al. showed at least a weak effect ($r > .099$) and two of those four showed a medium effect ($r > .299$). In terms of predicting text readability, six of the components developed in the current study explained ~44% of the variance with the strongest predictor the informational component (negative predictor) followed by the common syntactic construction component and the narrativity component (positive predictors). There were three other negative predictors that indicated reading ease was a function of less complex noun phrases, more difficult function words, and less lexical variety. The Graesser et al. components explained ~36% of the variance in the reading ease scores with the strongest predictors related to narrativity, syntactic similarity, and word concreteness.

Overall, the results indicate that the components derived from this study have strong overlap with theories of text difficulty and show strong predictive ability in a readability prediction task. The components also show overlap with many of the components derived in Graesser et al. (2011) providing some indication of convergent validity. Practically, these components should be easier to understand at a construct level than individual linguistic features. For instance, providing feedback to materials developers, teachers, administrators, or students about the difficulty of text in terms of a text being more strongly narrative or informational should make allow for greater understanding of the language constructs in the text that lead to comprehension. This is in contrast to discussing individual features that may inform these components such as orthographic neighborhoods, age of exposure, morpheme complexity, or verb construction faith. The same can be said about explaining language components such as lexical ease, lexical variety, and phrasal complexity, which have better accessibility in terms of interpretation.

CONCLUSION

This study has demonstrated the practicality of aggregating linguistic features into language components related to text readability from an analytic, theoretical, and practical perspective. Seven language components were developed in this study, all of which correlated with text readability and all of which overlapped with theoretical understandings of the reading process. The components were also strong predictors of text readability and could be integrated into a readability formula that performed well, was reliable, and allowed for interpretation. From a practical perspective, the components are easier to understand for non-researchers which would likely increase stake-holder participation. Thus, if integrated into a readability formula, it would not only be the output of the formula that was beneficial but also the interpretation of how the components influenced that output.

This study builds on Graesser et al. (2011) by introducing more advanced NLP features, better controlling for statistical assumptions, and assessing reading component performance on readability criteria. While, it has limitations, these can be built into future studies.

For instance, SALAT contains thousands of linguistic features related to many aspects of text readability, but it does not measure features related to rhetorical functions, pragmatics, or semantics (to name but a few). As NLP tools continue to advance and allow for the integration of additional language features, newer components should be developed to capture these features. Additionally, this study focused on the CLEAR corpus as the unit of analysis. While the CLEAR corpus is likely the largest readability corpus that contains unique readability criteria for each text excerpt, it only focuses on two genres of texts (informational and literary), oversamples from older texts, and only contains texts developed for elementary and secondary students. As larger corpora become available, text

patterns may change, and additional components related to discourse types may arise. This may help to increase the proportion of variance explained in the CLEAR corpus, which was relatively low in the current analysis (~50%).

Lastly, we used a principal component analysis (PCA) instead of an exploratory factor analysis (EFA). We selected a PCA because our interest was in data reduction with the end goal of developing a small set of components related to readability that contained information that was on par with the individual features usually found in readability formulas. The PCA approach also allows us to identify the items within the component and assess their strength within that component and within other components (Grimm & Yarnold, 1995). An EFA approach would allow researchers to better understand the correlations within the factors found. It would also allow for multi-dimensional analysis (MDA) approach such as those used by Biber and colleagues. An MDA would allow for an examination of how factors load on meta-data including genre of text, year of publication, and other text variables (Biber, 1988).

Note

1. It should be noted that the texts in the TASA corpus appear to be dependent (i.e., multiple excerpts that mention that same characters and events appear in the TASA corpus). An assumption of a PCA is independent data. Thus, the components developed in Graesser et al. may be problematic from a statistical perspective. However, there is little information about TASA and it is difficult to track down the original texts to double check.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

This paper makes use of the CommonLit Ease of Readability Corpus (CLEAR). The corpus is located here.

Ethics statement

The development of the CLEAR corpus and subsequent analyses of the data was approved by the Internal Review Board at Georgia State University. No data was collected from participants for this study.

References

Allington, R. L. (2005). NCLB, reading first, and whither the future. *Reading Today*, *23*(2), 18.

Berninger, V. B., Nagy, W. E., Carlisle, J. F., Thomson, J., Hoffer, D., Abbott, S., Abbott, R., Richards, T., & Aylward, E. (2003). Effective treatment for children with dyslexia in grades 4–6: Behavioral and brain evidence. In B. Foorman (Ed.), *Preventing and remediating reading difficulties: Bringing science to scale* (pp. 381–347). York Press.

Best, R., Floyd, R., & McNamara, D. (2008). Differential competencies contributing to children's comprehension of narrative and expository texts. *Reading Psychology*, *29*, 137–164. https://doi.org/10.1080/02702710801963951

Biber, D. (1988). *Variation across speech and writing*. Cambridge University Press. https://doi.org/10.1017/CBO9780511621024

Biber, D., Conrad, S., Reppen, R., Byrd, P., & Helt, M. (2002). Speaking and writing in the university: A multi-dimensional comparison. *TESOL Quarterly*, *36*(1), 9–48.

Bradley, M. M., & Lang, P. J. (1999). *Affective norms for English words (ANEW): Instruction manual and affective ratings. Technical report C-1*. The Center for Research in Psychophysiology, University of Florida.
Bradley, R. A., & Terry, M. E. (1952). Rank analysis of incomplete block designs. I. The method of paired comparisons. *Biometrika*, *39*(3–4), 324–345. https://doi.org/10.1093/biomet/39.3-4.324
Brezina, V., & Pallotti, G. (2019). Morphological complexity in written L2 texts. *Second Language Research*, *35*(1), 99–119. https://doi.org/10.1177/0267658316643125
Brysbaert, M., & New, B. (2009). Moving beyond Kučera and Francis: A critical evaluation of current word frequency norms and the introduction of a new and improved word frequency measure for American English. *Behavior Research Methods*, *41*(4), 977–990. https://doi.org/10.3758/BRM.41.4.977
Brysbaert, M., Warriner, A. B., & Kuperman, V. (2014). Concreteness ratings for 40 thousand generally known English word lemmas. *Behavior Research Methods*, *46*(3), 904–911. https://doi.org/10.3758/s13428-013-0403-5
Chall, J. S., & Dale, E. (1995). *Readability revisited: The new dale-chall readability formula*. Brookline Books.
Collins-Thompson, K. (2014). Computational assessment of text readability: A survey of current and future research. *ITL - International Journal of Applied Linguistics*, *165*(2), 97–135. https://doi.org/10.1075/itl.165.2.01col
Coltheart, M. (1981). The MRC psycholinguistic database. *The Quarterly Journal of Experimental Psychology Section A*, *33*(4), 497–505. https://doi.org/10.1080/14640748108400805
Common Core State Standards Initiative. (2010, June). *Common core state standards for English language arts & literacy in history/social studies, science and technical subjects*. CCSSO & National Governors Association.
Costello, A. B., & Osborne, J. (2005). Best practices in exploratory factor analysis: Four recommendations for getting the most from your analysis. *Practical Assessment, Research & Evaluation*, *10*(7), 1–9.
Crossley, S. A., Heintz, A., Choi, J., Batchelor, J., Karimi, M., & Malatinszky, A. (2023). A large-scaled corpus for assessing text readability. *Behavior Research Methods*, *55*(2), 491–507. https://doi.org/10.3758/s13428-022-01802-x
Crossley, S. A., & Kyle, K. (2018). Analyzing spoken and written discourse: A role for natural language processing tools. In A. Phakiti, P. De Costa, L. Plonsky, & S. Starfield, (Eds.), *The Palgrave Handbook of Applied Linguistics Research Methodology*. Palgrave Macmillan. https://doi.org/10.1057/978-1-137-59900-1_25
Crossley, S. A., Kyle, K., & Dascalu, M. (2019). The tool for the automatic analysis of cohesion 2.0: Integrating semantic similarity and text overlap. *Behavior Research Methods*, *51*(1), 14–27. https://doi.org/10.3758/s13428-018-1142-4
Crossley, S. A., Kyle, K., & McNamara, D. S. (2016). The tool for the automatic analysis of text cohesion (TAACO): Automatic assessment of local, global, and text cohesion. *Behavior Research Methods*, *48*(4), 1227–1237. https://doi.org/10.3758/s13428-015-0651-7
Crossley, S. A., Kyle, K., & McNamara, D. S. (2017). Sentiment analysis and social cognition engine (SEANCE): An automatic tool for sentiment, social cognition, and social order analysis. *Behavior Research Methods*, *49*(3), 803–821. https://doi.org/10.3758/s13428-016-0743-z
Crossley, S. A., Skalicky, S., & Dascalu, M. (2019). Moving beyond classic readability formulas: New methods and new models. *Journal of Research in Reading*, *42*(3–4), 541–561. https://doi.org/10.1111/1467-9817.12283
Crossley, S. A., Skalicky, S., Dascalu, M., McNamara, D. S., & Kyle, K. (2017). Predicting text comprehension, processing, and familiarity in adult readers: New approaches to readability formulas. *Discourse Processes*, *54*(5–6), 340–359. https://doi.org/10.1080/0163853X.2017.1296264
Crossley, S. A., Tywoniw, R., & Choi, J. S. (2024). The tool for automatic measurement of morphological information (TAMMI). *Behavior Research Methods*, *56*, 5918–5929. https://doi.org/10.3758/s13428-023-02324-w
Cunningham, J. W., Spadorcia, S. A., Erickson, K. A., Koppenhaver, D. A., Sturm, J. M., & Yoder, D. E. (2005). Investigating the instructional supportiveness of leveled texts. *Reading Research Quarterly*, *40*(4), 410–427. https://doi.org/10.1598/RRQ.40.4.2
Davies, M. (2010). The corpus of contemporary American English as the first reliable monitor corpus of English. *Literary and Linguistic Computing*, *25*(4), 447–464. https://doi.org/10.1093/llc/fqq018
De Clercq, O., Hoste, V., Desmet, B., Van Oosten, P., De Cock, M., & Macken, L. (2014). Using the crowd for readability prediction. *Natural Language Engineering*, *20*(3), 293–325. https://doi.org/10.1017/S1351324912000344
Dufty, D. F., Graesser, A. C., Louwerse, M. M., & McNamara, D. S. (2006). Assigning grade levels to textbooks: Is it just readability? *Proceedings of the Annual Meeting of the Cognitive Science Society*, *28*(28).
Flesch, R. (1948). A new readability yardstick. *Journal of Applied Psychology*, *32*(3), 221–233. https://doi.org/10.1037/h0057532
Gernsbacher, M. A. (1990). *Language comprehension as structure building*. Erlbaum.
Givón, T. (1995). *Functionalism and grammar*. John Benjamins.
Goulart, L., & Wood, M. (2019). Methodological synthesis of research using multi-dimensional analysis. *Journal of Research Design and Statistics in Linguistics and Communication Science*, *6*(2). https://doi.org/10.1558/jrds.18454
Graesser, A. C., McNamara, D. S., Cai, Z., Conley, M., Li, H., & Pennebaker, J. (2014). Coh-Metrix measures text characteristics at multiple levels of language and discourse. *The Elementary School Journal*, *115*(2), 210–229. https://doi.org/10.1086/678293
Graesser, A. C., McNamara, D. S., & Kulikowich, J. M. (2011). Coh-Metrix: Providing multilevel analyses of text characteristics. *Educational Researcher*, *40*(5), 223–234. https://doi.org/10.3102/0013189X11413260

Graesser, A. C., McNamara, D. S., Louwerse, M. M., & Cai, Z. (2004). Coh-Metrix: Analysis of text on cohesion and language. *Behavioral Research Methods, Instruments, and Computers*, *36*(2), 193–202. https://doi.org/10.3758/BF03195564

Grimm, L. G., & Yarnold, P. R. (1995). *Reading and understanding multivariate statistics*. American psychological association.

Grömping, U. (2006). Relative importance for linear regression in R: The package relaimpo. *Journal of Statistical Software*, *17*(1), 1–27. https://doi.org/10.18637/jss.v017.i01

Halliday, M. A. K., & Hasan, R. (1976). *Cohesion in English*. Routledge.

Hoffman, P., Ralph, M. A. L., & Rogers, T. T. (2013). Semantic diversity: A measure of semantic ambiguity based on variability in the contextual usage of words. *Behavior Research Methods*, *45*(3), 718–730. https://doi.org/10.3758/s13428-012-0278-x

Howes, D. H., & Solomon, R. L. (1951). Visual duration thresholds as a function of word probability. *Journal of Experimental Psychology*, *41*(6), 401–410. https://doi.org/10.1037/h0056020

Jackson, G. T., Allen, L. K., & McNamara, D. S. (2016). Common core TERA: Text ease and readability assessor. In S. A. Crossley & D. S. McNamara (Eds.), *Adaptive educational technologies for literacy instruction* (pp. 49–68). Routledge.

Jarvis, S. (2013). Capturing the diversity in lexical diversity. *Language Learning*, *63*(S1), 87–106. https://doi.org/10.1111/j.1467-9922.2012.00739.x

Johnson, W. (1944). Studies in language behavior: I. A program of research. *Psychological Monographs*, *56*(2), 1–15. https://doi.org/10.1037/h0093508

Juel, C., & Solso, R. L. (1981). The role of orthographic redundancy, versatility and spelling-sound correspondences in word identification. In M. L. Kamil (Ed.), *Directions in reading: Research and instruction* (pp. 74–82). National Reading Conference.

Just, M. A., & Carpenter, P. A. (1980). A theory of reading: From eye fixations to comprehension. *Psychological Review*, *87*(4), 329–354. https://doi.org/10.1037/0033-295X.87.4.329

Kaiser, H. F., & Rice, J. (1974). Little jiffy, mark IV. *Educational and Psychological Measurement*, *34*(1), 111–117. https://doi.org/10.1177/001316447403400115

Kincaid, J. P., Fishburne, R. P., Rogers, R. L., & Chissom, B. S. (1975). Derivation of new readability formulas (automated readability index, fog count, and flesch reading ease formula) for navy enlisted personnel. *Research Branch Report*, 8–75. Chief of Naval Technical Training: Naval Air Station Memphis.

Kiss, G. R., Armstrong, C., Milroy, R., & Piper, J. (1973). An associative thesaurus of English and its computer analysis. In A. J. Aitken, R. W. Bailey, & N. Hamilton-Smith (Eds.), *The computer and literary studies* (pp. 153–165). Edinburgh University Press.

Klare, G. R. (1952). Measures of the readability of written communication: An evaluation. *Journal of Educational Psychology*, *43*(7), 385–399. https://doi.org/10.1037/h0058972

Koslin, B. I., Zeno, S., & Koslin, S. (1987). *The DRP: An effective measure in reading*. College Entrance Examination Board.

Kuhn, M. (2008). Building predictive models in R using the caret package. *Journal of Statistical Software*, *28*(5), 1–26. https://doi.org/10.18637/jss.v028.i05

Kuperman, V., Stadthagen-Gonzalez, H., & Brysbaert, M. (2012). Age-of-acquisition ratings for 30,000 English words. *Behavior Research Methods*, *44*(4), 978–990. https://doi.org/10.3758/s13428-012-0210-4

Kyle, K. (2016). *Measuring syntactic development in L2 writing: Fine grained indices of syntactic complexity and usage-based indices of syntactic sophistication* [Doctoral Dissertation]. http://scholarworks.gsu.edu/alesl_diss/35

Kyle, K., & Crossley, S. (2017). Assessing syntactic sophistication in L2 writing: A usage-based approach. *Language Testing*, *34*(4), 513–535. https://doi.org/10.1177/0265532217712554

Kyle, K., & Crossley, S. A. (2015). Automatically assessing lexical sophistication: Indices, tools, findings, and application. *TESOL Quarterly*, *49*(4), 757–786. https://doi.org/10.1002/tesq.194

Kyle, K., Crossley, S. A., & Berger, C. (2018). The tool for the automatic analysis of lexical sophistication (TAALES): Version 2.0. *Behavior Research Methods*, *50*(3), 1030–1046. https://doi.org/10.3758/s13428-017-0924-4

Kyle, K., Crossley, S. A., & Jarvis, S. (2020). Assessing the validity of lexical diversity indices using direct judgements. *Language Assessment Quarterly*, *18*(2), 1–17. https://doi.org/10.1080/15434303.2020.1844205

Landauer, T. K., Foltz, P. W., & Laham, D. (1998). An introduction to latent semantic analysis. *Discourse Processes*, *25*(2–3), 259–284. https://doi.org/10.1080/01638539809545028

Lindeman, R. H., Merenda, P. F., & Gold, R. Z. (1980). *Introduction to Bivariate and multivariate analysis*. Foresman.

Lund, K., & Burgess, C. (1996). Producing high-dimensional semantic spaces from lexical co-occurrence. *Behavior Research Methods Instruments & Computers*, *28*(2), 203–208. https://doi.org/10.3758/BF03204766

McCarthy, P. M., & Jarvis, S. (2010). MTLD, vocd-d, and HD-D: A validation study of sophisticated approaches to lexical diversity assessment. *Behavior Research Methods*, *42*(2), 381–392. https://doi.org/10.3758/BRM.42.2.381

McDonald, S. A., & Shillcock, R. C. (2001). Rethinking the word frequency effect: The neglected role of distributional information in lexical processing. *Language and Speech*, *44*(3), 295–323. https://doi.org/10.1177/00238309010440030101

McNamara, D. S., Graesser, A. C., McCarthy, P. M., & Cai, Z. (2014). *Automated evaluation of text and discourse with Coh-Metrix*. Cambridge University Press.
McNamara, D. S., & Magliano, J. (2009). Toward a comprehensive model of comprehension. *Psychology of Learning and Motivation*, *51*, 297–384. https://doi.org/10.1016/S0079-7421(09)51009-2
Mesmer, H. A. (2005). Decodable text and the first grade reader. *Reading & Writing Quarterly*, *21*(1), 61–86. https://doi.org/10.1080/10573560590523667
Mesmer, H. A., Cunningham, J. W., & Hiebert, E. H. (2012). Toward a theoretical model of text complexity for the early grades: Learning from the past, anticipating the future. *Reading Research Quarterly*, *47*(3), 235–258. https://doi.org/10.1002/rrq.019
Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word–emotion association lexicon. *Computational Intelligence*, *29*(3), 436–465. https://doi.org/10.1111/j.1467-8640.2012.00460.x
Nagy, W. E., & Anderson, R. (1984). How many words are there in printed school English? *Reading Research Quarterly*, *19*(3), 304–330. https://doi.org/10.2307/747823
O'Reilly, T., Wang, Z., & Sabatini, J. (2019). How much knowledge is too little? When a lack of knowledge becomes a barrier to comprehension. *Psychological Science*, *30*(9), 1344–1351. https://doi.org/10.1177/0956797619862276
Osgood, C., Suci, G., & Tannenbaum, P. (1957). *The measurement of meaning*. University of Illinois.
R Core Team. (2021). *R: A language and environment for statistical computing*. URL R Foundation for Statistical Computing. https://www.R-project.org/
Revelle, W. (2023). *Psych: Procedures for psychological, psychometric, and personality research*. R package version 2.3.6, Northwestern University. https://CRAN.R-project.org/package=psych
Richardson, J. T. E. (1975). The effect of word imageability in acquired dyslexia. *Neuropsychologia*, *13*(3), 281–288. https://doi.org/10.1016/0028-3932(75)90004-4
Sáenz, L. M., & Fuchs, L. S. (2002). Examining the reading difficulty of secondary students with learning disabilities: Expository versus narrative text. *Remedial and Special Education*, *23*(1), 31–41. https://doi.org/10.1177/074193250202300105
Sánchez-Gutiérrez, C. H., Mailhot, H., Deacon, S. H., & Wilson, M. A. (2018). MorphoLex: A derivational morphological database for 70,000 English words. *Behavior Research Methods*, *50*(4), 1568–1580. https://doi.org/10.3758/s13428-017-0981-8
Sheehan, K. M., Flor, M., Napolitano, D., & Ramineni, C. (2015). Using TextEvaluator ® to quantify sources of linguistic complexity in textbooks targeted at first-grade readers over the past half century. *ETS Research Report Series*, *2015*(2), 1–20. https://doi.org/10.1002/ets2.12085
Sheehan, K. M., Kostin, I., Napolitano, D., & Flor, M. (2014). The TextEvaluator tool: Helping teachers and test developers select texts for use in instruction and assessment. *The Elementary School Journal*, *115*(2), 184–209. https://doi.org/10.1086/678294
Smith, D., Stenner, A. J., Horabin, I., & Smith, M. (1989). *The Lexile scale in theory and practice: Final report*. MetaMetrics.
Snow, C. (Ed.). (2002). *Reading for understanding: Toward an R & D program in reading comprehension*. Rand.
Stanovich, K. E. (1985). Explaining the variance in reading ability in terms of psychological processes: What have we learned? *Annals of Dyslexia*, *35*(1), 67. https://doi.org/10.1007/BF02659181
Stone, P., Dunphy, D. C., Smith, M. S., Ogilvie, D. M., & associates. (1966). *The General inquirer: A computer approach to content analysis*. The MIT Press.
Turner, H., & Firth, D. (2020). *Bradley-terry models* (version 1.1-2). https://github.com/hturner/BradleyTerry2
Wang, Z., Sabatini, J., O'Reilly, T., & Feng, G. (2017). How individual differences interact with task demands in text processing. *Scientific Studies of Reading*, *21*(2), 165–178. https://doi.org/10.1080/10888438.2016.1276184
Wolfe, M. B., Schreiner, M. E., Rehder, B., Laham, D., Foltz, P. W., Kintsch, W., & Landauer, T. K. (1998). Learning from text: Matching readers and texts by latent semantic analysis. *Discourse Processes*, *25*(2–3), 309–336. https://doi.org/10.1080/01638539809545030